

PRISM — Final Stack Summary & Cost Calculation

1. The architecture we landed on

Hybrid tiered system (not pure LLM, not pure local — a layered approach):

Layer	What it does	Powered by
Bulk layer (every article)	Source link + outlet metadata + bias/credibility score	Classical ML classifier (TF-IDF/logistic regression → later DistilBERT) — trained once, runs cheap forever
Default verification	Multi-agent debate on flagged/trending stories	Free-tier open model (Groq/Gemini) or local model via Ollama for dev
Premium/Pro verification	Full multi-agent debate, deepest quality	Claude API — reserved for paying users or high-stakes flagged stories
Caching	Same story verified once, served to everyone who asks	Upstash (free tier)

This is the structure that makes the cost math below actually work — cost scales with real usage and tiering, not with raw article volume.

2. Full cost table

Item	Cost	Notes
Claude (you, as engineer/dev assistant)	\$0	Free access you already have
Domain	\$0	Free access you already have
Hosting (Vercel/Railway)	\$0	Free tier
Database (Supabase)	\$0	Free tier
Auth (Clerk)	\$0	Free tier
Caching (Upstash)	\$0	Free tier
Error monitoring (Sentry)	\$0	Free tier
Email (Resend)	\$0	Free tier
Analytics (PostHog)	\$0	Free tier
Vector DB (Pinecone)	\$0	Free tier
Classical ML bias/credibility scoring	\$0	Runs on CPU, no per-call cost
Bulk multi-agent verification (Groq/Gemini free tier)	\$0 (rate-limited)	Free quota; breaks down only at high traffic
Local dev/testing model (Ollama)	\$0	Your machine, dev only
Premium Claude-powered verification (Pro tier / flagged stories only)	~\$0.01-0.03 per verification	Only real variable cost
Stripe	2.9% + \$0.30	Only charged when you earn revenue

3. Realistic monthly total, by stage

Stage	Verify- clicks/month	Estimated real cost
Pre-launch / testing	~0	\$0
Early users (friends, campus, first real testers)	~500	\$0-5 (mostly absorbed by free tiers + caching)
Small real traction	~5,000	~\$15-40 (some spill into paid Claude tier for Pro users)
Meaningful traction	~20,000+	~\$50-150 (this is the point revenue should already be covering it via Pro subscriptions)

Bottom line: PRISM can realistically launch and run for \$0/month while you build and test, and stay under ~\$5-15/month through early real usage — as long as the bulk layer stays on classical ML + free-tier models, and Claude-quality verification is reserved for Pro users or genuinely high-stakes stories.

4. The one thing worth remembering

None of this is "fully free forever" if PRISM actually succeeds — and that's fine. **Cost only grows when usage grows, and usage growing is the good outcome.** The design goal was never zero cost at scale; it was zero cost *risk* while you're solo, unfunded, and still validating the idea. That's exactly what this stack achieves.

Ready when you want to move into the actual Phase 0 build checklist (real ingestion pipeline, provenance fixes, and wiring up this tiered verification system).